CHE 201 HW 8

Due: 11.59pm on 04/19/2020

1. (10 points)

5.3 Classify the following processes of a closed system as *possible, impossible*, or *indeterminate*.

	Entropy Change	Entropy Transfer	Entropy Production
(a)	>0	0	
(b)	<0		>0
(c)	0	>0	
(d)	>0	>0	
(e)	0	<0	
(f)	>0		<0
(g)	<0	<0	

2. (15 points)

5.5 As shown in Fig. P5.5, a reversible power cycle R and an irreversible power cycle I operate between the same hot and cold thermal reservoirs. Cycle I has a thermal efficiency equal to one-third of the thermal efficiency of cycle R.

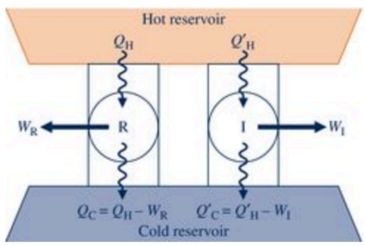

FIGURE P5.5

a. If each cycle receives the same amount of energy by heat transfer from the hot reservoir, determine which cycle (i) develops greater net work, (ii) discharges greater energy by heat transfer to the cold reservoir.

b. If each cycle develops the same net work, determine which cycle (i) receives greater energy by heat transfer from the hot reservoir, (ii) discharges greater energy by heat transfer to the cold reservoir.

3. (15 points)

5.10 Figure P5.10 shows two power cycles, denoted 1 and 2, operating in series, together with three thermal reservoirs. The energy transfer by heat into cycle 2 is equal in magnitude to the energy transfer by heat from cycle 1. All energy transfers are positive in the directions of the arrows.

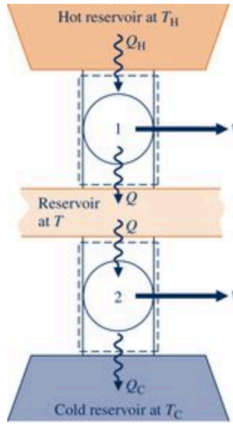

FIGURE P5.10

a. Determine an expression for the thermal efficiency of an overall cycle consisting of cycles 1 and 2 expressed in terms of their individual thermal efficiencies.

b. If cycles 1 and 2 are each reversible, use the result of part (a) to obtain an expression for the thermal efficiency of the overall cycle in terms of the temperatures of the three reservoirs, T_H, T, and T_C, as required. Comment.

c. If cycles 1 and 2 are each reversible and have the same thermal efficiency, obtain an expression for the intermediate temperature T in terms of T_H and T_C.

4. (15 points)

5.14 A power cycle receives Q_H by heat transfer from a hot reservoir at T_H = 1200 K and rejects energy Q_C by heat transfer to a cold reservoir at T_C = 400 K. For each of the following cases, determine whether the cycle operates *reversibly*, operates *irreversibly*, or is *impossible*.

a. Q_H = 900 kJ, W_{cycle} = 450 kJ

b. Q_H = 900 kJ, Q_C = 300 kJ

c. W_{cycle} = 600 kJ, Q_C = 400 kJ

d. η = 75%

5. (15 points)

5.22 WP As shown in Fig. P5.22, two reversible cycles arranged in series each produce the same net work, W_{cycle}. The first cycle receives energy Q_H by heat transfer from a hot reservoir at 1000°R and rejects energy Q by heat transfer to a reservoir at an intermediate temperature, T. The second cycle receives energy Q by heat transfer from the reservoir at temperature T and rejects energy Q_C by heat transfer to a reservoir at 400°R. All energy transfers are positive in the directions of the arrows. Determine

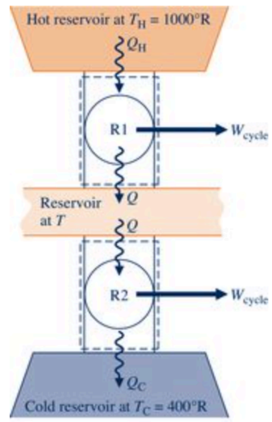

FIGURE P5.22

a. the intermediate temperature T, in °R, and the thermal efficiency for each of the two power cycles.

b. the thermal efficiency of a *single* reversible power cycle operating between hot and cold reservoirs at 1000°R and 400°R, respectively. Also, determine the net work developed by the single cycle, expressed in terms of the net work developed by each of the two cycles, W_{cycle}.

6. (15 points)

5.44 WP A heat pump cycle is used to maintain the interior of a building at 21°C. At steady state, the heat pump receives energy by heat transfer from well water at 9°C and discharges energy by heat transfer to the building at a rate of 120,000 kJ/h. Over a period of 14 days, an electric meter records that 1490 kW · h of electricity is provided to the heat pump. Determine

a. the amount of energy that the heat pump receives over the 14-day period from the well water by heat transfer, in kJ.

b. the heat pump's coefficient of performance.

c. the coefficient of performance of a reversible heat pump cycle operating between hot and cold reservoirs at 21°C and 9°C.

7. (15 points)

5.60 WP Figure P5.60 gives the schematic of a vapor power plant in which water steadily circulates through the four components shown. The water flows through the boiler and condenser at constant pressure and through the turbine and pump adiabatically. Kinetic and potential energy effects can be ignored. Process data follow:

Process 4–1: constant pressure at 1 MPa from saturated liquid to saturated vapor

Process 2–3: constant pressure at 20 kPa from x_2 = 88% to x_3 = 18%

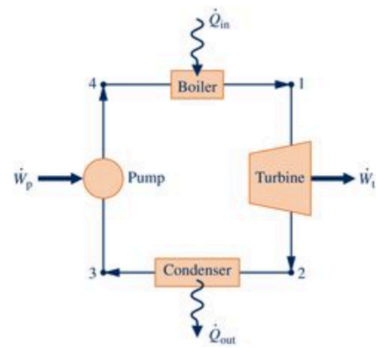

FIGURE P5.60–62

a. Using Eq. 5.13 expressed on a time-rate basis, determine if the cycle is *internally reversible, irreversible*, or *impossible.*

b. Determine the thermal efficiency using Eq. 5.4 expressed on a time-rate basis and steam table data.

c. Compare the result of part (b) with the *Carnot efficiency* calculated using Eq. 5.9 with the boiler and condenser temperatures and comment.